

VAYU-RAKSHA

Anticipatory Cyclone & Infrastructure Cascade Intelligence Platform

"From satellite to autonomous municipal action — 72 hours before landfall"

THE GAP WE FILL

Every existing solution stops at ADVISORY GENERATION (what will happen).
None models how failures CASCADE across interdependent infrastructure networks,
or optimizes PRE-LANDFALL hardening. VAYU-RAKSHA is the first 5-agent system
to simulate cascade failure propagation and fuse ISRO MOSDAC + RISAT-1A with GEE.

BHUMI Agent

VAYU Agent

SETU Agent

SANCHAR Agent

NIRNAY (Supervisor)

ISRO MOSDAC + RISAT-1A | Free Open Data

Google Earth Engine | Gemini 3.7 Flash

[CONTEXT]

VAYU-RAKSHA

Anticipatory Cyclone & Infrastructure Cascade Intelligence Platform

Research: What Existing Solutions Do
"From satellite to autonomous municipal action — 72 hours before landfall"

THE GAP WE FILL		
Feature	Other Projects (8+ reviewed)	VAYU-RAKSHA
Cyclone Track + Wind Field	Holland vortex / IBTrACS — everyone does this	Same base + MOSDAC INSAT-3DS validation
Storm Surge Modeling	Bathtub / SLOSH proxy — static maps only	Physics-informed + time-stepped hourly surge
Infrastructure Risk	Simple spatial intersection of GEE layers	Full NetworkX cascade failure graph
Cross-domain Cascade Failure	NOT DONE by any reviewed project	Power → Hospital → Comms → Roads chain
ISRO Data Integration	GEE only (Sentinel/NASA) — no ISRO native	MOSDAC + RISAT-1A + INSAT-3DS fused
Multi-Agent Architecture	Single Gemini LLM with tools — not true agents	5 specialized LangGraph agents + supervisor
Pre-landfall Action Protocol	Advisory text only (post-fact suggestions)	Ranked autonomous hardening action queue
Counterfactual Scenario Engine	NOT in any reviewed project	"If we de-energize X now — lives saved?"
Parametric Insurance Trigger	Mentioned/simulated — no real logic	GEE-validated threshold + auto trigger event
Multilingual Comms	Text translation only (3-4 languages)	Context-aware SMS + IVR (6 languages)

KEY INSIGHT: Every existing project solves the OBSERVATION problem (what will happen) but none solves the ACTION problem (what must be done NOW, in what sequence, to minimize cascade damage). VAYU-RAKSHA fills this critical gap with a cascade-aware, multi-agent action optimizer.

[INNOVATION]

VAYU-RAKSHA

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THE GAP WE FILL

Every existing system
None model cascading
or optimized for
to simulate

Layer	Node Type	Dependency Chain	VAYU-RAKSHA Action
L1	Transmission Lines 220kV Substations	→ Hospital Power → Shelter Power	Identify + pre-isolate
L2	Hospitals (Trauma & Maternity)	→ ICU equipment → Cold chain vaccines	Fuel stockpile advisory
L3	Arterial Roads & Bridges	→ Evacuation routes → Relief convoy paths	Flood-penalized rerouting
L4	Telecom Towers & Control Rooms	→ Early warning reach → NDRF coordination	Backup link activation
L5	Water Treatment & Pumping Stations	→ Hospital supply → Shelter sanitation	Reserve tank advisory

BHUMI Agent

NIRNAY (Supervisor)

ISRO MOSDAC

2. ISRO MOSDAC + RISAT-1A Native Integration

3. Counterfactual Pre-Landfall Action Optimizer

4. True 5-Agent LangGraph Architecture

TEAM: SYNTRIX | Build with AI — Code for Communities (2nd Edition)

Tech Stack: Python · FastAPI · LangGraph · Google Earth Engine · Gemini 3.7 Flash · NetworkX · React

Data: ISRO MOSDAC · INSAT-3DS · RISAT-1A SAR · Sentinel-1 SAR · GPM IMERG · IMD · OpenStreetMap

github.com/Niss54/vayu-raksha

[ARCHITECTURE]

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DATA INGESTION LAYER	ISRO MOSDAC (INSAT-3DS, Oceansat-3, RISAT-1A) · GEE (Sentinel-1 SAR, NAS
↓	
AGENT ORCHESTRATION (LangGraph S	Supervisor agent (NIRNAY) manages 4 specialized agents in parallel and sequential
↓	
PROCESSING ENGINES	• Holland Wind Field Model · GEE Storm Surge (physics-informed bathtub + hydrody
↓	
ACTION & OUTPUT LAYER	Pre-landfall action queue (ranked by cascade impact) · Municipal PDF/WhatsApp ad

Optimizes PRE-LANDFALL hardening. VAYU-RAKSHA is the first agent system to simulate cascade failure propagation and fuse ISRO MOSDAC + RISAT-1A with GEE.

T-72h GENESIS	Day 1 AM	BHUMI Agent: GEE terrain + infrastructure layer build VAYU Agent: Initial cyclone track + intensity forecast ingested from IMD/ECMWF SETU Agent: Cascade failure graph initialized with OSM node data	Supervisor)
T-48h APPROACH	Day 1 PM	VAYU Agent: Holland wind field generated, surge corridors identified SETU Agent: Infrastructure risk scores computed, cascade chains identified NIRNAY Supervisor: Counterfactual optimizer runs — pre-landfall action queue generated	
T-24h PRE-LAND	Day 2 AM	SANCHAR Agent: District-level advisories dispatched (6 languages) NIRNAY: Action priorities sent to DM authorities (substation isolation, route alerts) Parametric insurance trigger conditions set and monitored	
T-6h LAND-FALL	Day 2 PM	All agents update on 1-hour cycle SAR satellite pass (RISAT-1A) — real-time flood inundation fed to SETU Evacuation route guardian re-runs with live flood data	
T+12h POST LAND	Day 3	Sentinel-1 SAR post-event flood extent map via GEE Parametric insurance trigger evaluated against GEE thresholds — auto-payout event Recovery sequence advisory generated by SANCHAR Agent	

[AGENTS]

VAYU-RAKSHA

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■ NIRNAY — Supervisor Agent

Orchestrator, Counterfactual Optimizer, Action Prioritizer

INPUTS	Outputs from all 4 sub-agents (risk scores, cascade graph, advisories, surge data)
OUTPUTS	Ranked pre-landfall action queue · Counterfactual scenario comparison · Prioritized DM brief · Final situation report
AI MODEL	Gemini 3.7 Flash (extended thinking mode) — multi-document reasoning across all agent outputs

■ ■ BHUMI — Earth Intelligence Agent

Satellite Data Ingestion & Terrain Modeling

INPUTS	GEE Python API (Sentinel-1 SAR, NASADEM DEM, JRC Surface Water, WorldPop, Dynamic World) · ISRO RISAT-1A SAR · MOSDAC INSAT-3DS
OUTPUTS	30m resolution terrain model · Pre-computed inundation susceptibility raster · Infrastructure exposure layer · SAR-derived flood masks (through monsoon cloud cover)
AI MODEL	GEE Compute (Python) + Gemini Vision for SAR image interpretation

■ VAYU — Atmospheric Intelligence Agent

Cyclone Track, Wind Field & Surge Simulation

INPUTS	IMD NWP track + intensity (6-hourly) · ECMWF ERA5 reanalysis · IBTrACS historical storm database · GPM IMERG real-time rainfall · MOSDAC cyclone forecast products
OUTPUTS	Holland parametric wind field (hourly snapshots) · Storm surge corridors (physics-informed bathtub model) · 72h rainfall accumulation grid · Intensity probability cone
AI MODEL	Gemini 3.7 Flash for MOSDAC bulletin interpretation · NumPy/SciPy for wind modeling

■ SETU — Infrastructure Cascade Agent

Dependency Graph, Cascade Failure & Evacuation Routing

TEAM: SYNTRIX | Build with AI — Code for Communities (2nd Edition)

Tech Stack: Python · FastAPI · LangGraph · Google Earth Engine · Gemini 3.7 Flash · NetworkX · React

Data: ISRO MOSDAC · INSAT-3DS · RISAT-1A SAR · Sentinel-1 SAR · GPM IMERG · IMD · OpenStreetMap

github.com/Niss54/vayu-raksha

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Anticipatory Systems & Infrastructure

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INPUTS	OSM Overpass API (roads, hospitals, substations, shelters, telecom) · BHUMI terrain layer · VAYU surge corridors · India GIS district boundaries
OUTPUTS	NetworkX interdependency graph with cascade failure propagation · Risk-ranked asset triage queue · Flood-penalized Dijkstra evacuation routes · Counterfactual impact per action (lives protected vs disruption cost)
AI MODEL	NetworkX graph algorithms + Shapely spatial ops + Gemini for narrative cascade analysis

■ SANCHAR — Communication & Advisory Agent	
Multilingual Advisory & Parametric Insurance Dispatch	
INPUTS	SETU cascade scores · NIRNAY action queue · Gemini language generation · MOSDAC intensity data
OUTPUTS	Multilingual SMS bulletins (Hindi, Odia, Bengali, Telugu, Tamil, English) · District Collector PDF brief · NDRF coordination message · Parametric insurance trigger event JSON · IVR audio script (gTTS) · WhatsApp-formatted advisory
AI MODEL	Gemini 3.7 Flash (multilingual prompting) + gTTS for audio + Python templating

to simulate cascade failure propagation and fuse ISRO MOSDAC + RISAT-1A with GEE.

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ISRO MOSDAC + RISAT-1A | Free Open Data

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[DATA & TECH]

VAYU-RAKSHA

Anticipatory Cyclone & Infrastructure

Cas	Data Source	Provider	What We Use	Access
	ISRO MOSDAC	ISRO SAC	INSAT-3DS cloud motion, cyclone intensity, Dvorak technique products, real-time 30-min updates	FREE
"From	ISRO RISAT-1A SAR	ISRO/Bhoonidhi	C-band SAR through-cloud flood mapping, 3m resolution, disaster mode activation	FREE
T	ISRO Oceansat-3	ISRO SAC	Sea surface temperature, wind vectors over Bay of Bengal, Ku-band scatterometer	FREE
E	Google Earth Engine	Google/ESA/NASA	Sentinel-1 SAR, NASADEM 30m DEM, GPM IMERG rainfall, JRC Water, WorldPop, Dynamic World	FREE
N	IMD Open Data	MoES/IMD	NWP cyclone track, intensity, 6-hourly bulletins, rainfall forecast, storm warning zones	FREE
or	IBTrACS North Indian Ocean	NOAA	Historical storm tracks + wind + pressure records covering Bay of Bengal (1842-present)	FREE
to	OpenStreetMap (OSM)	OSM/Overpass	Roads, hospitals, substations, shelters, bridges, telecom towers — India coverage	FREE
	Copernicus Emergency	EU Copernicus	Post-event validated flood maps for model calibration	FREE

Technology Stack

AI / LLM	Gemini 3.7 Flash (multimodal + extended thinking) via Vertex AI
Agent Framework	LangGraph StateGraph (5 nodes) · LangChain tools · AutoGen sub-task spawning
Geospatial Core	Google Earth Engine Python API · GeoPandas · Shapely · Rasterio · Folium
Graph & Routing	NetworkX (cascade failure graph) · OSMnx (road network) · SciPy (spatial indexing)
Physics Modeling	NumPy/SciPy (Holland wind field, bathtub surge) · Custom hydrodynamic proxy
Backend API	FastAPI · Python 3.11 · Pydantic v2 · asyncio · WebSocket for live updates
Frontend / Viz	React 18 · Mapbox GL JS / Leaflet · Recharts · Tailwind CSS
Infrastructure	Google Cloud Run · Firebase Realtime DB · Google Cloud Storage · Docker
Advisory / Comms	Gemini multilingual prompting · gTTS (IVR audio) · Jinja2 PDF templates
Dev Tools	Poetry · pytest · GitHub Actions CI/CD · pre-commit

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github.com/Niss54/vayu-raksha

[CORE INNOVATION]

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THE GAP WE FILL

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STEP 1	VAYU Agent outputs hourly flood probability per grid cell at T-72h to T+24h
STEP 2	BHUMI Agent overlays infrastructure nodes on surge probability — assigns P(flood) to each node
STEP 3	SETU Agent iterates cascade: nodes where P(flood) > 0.7 are marked FAILED; all downstream dependents receive
STEP 4	Population affected = sum of WorldPop cells served by failed nodes; cascade depth tracked up to 3 hops
STEP 5	Counterfactual: For each node N in top-10 cascade initiators — simulate 'what if N is hardened/isolated pre-landfall?
STEP 6	NIRNAY ranks hardening actions by (delta-population-protected / hardening-cost-proxy) and outputs ranked action q

Sample Output — Ranked Action Queue

Priority	Action	Assets Protected	Cascade Nodes Saved	Window
#1 CRITICAL	De-energize coastal substations S-7, S-12, S-19	3 Hospitals, 2 Shelters, 1 Desalination	4 downstream	T-36h
#2 HIGH	Pre-position fuel tankers at Hospital H-3, H-8	2 ICUs, Blood banks, OT suites	3 downstream	T-48h
#3 HIGH	Close bridge RB-22 (surge threshold: 2.1m expected)	1 school, 1 fire station	N/A	T-12h
#4 MED	Activate generator backup: Telecom towers T-10, T-15	100% call coordination	6 downstream	T-24h
#5 MED	Pre-fill emergency water reserves: WP-2, WP-5	2 hospital wings, 4 shelters	5 downstream	T-48h

PARAMETRIC INSURANCE INTEGRATION: When VAYU Agent detects sustained wind speed ≥ 89 km/h within 50km of a GEE-defined grid cell AND Sentinel-1 SAR confirms flood extent $\geq 30\%$ in that cell — SANCHAR Agent generates a parametric trigger event JSON dispatched to insurer API. No loss adjustment required. Payout initiates within hours, not weeks.

[SATELLITE INTELLIGENCE]

VAYU-RAKSHA

Anticipatory Cyclone & Infrastructure Cascade Intelligence Platform

"From Data to Decision: Anticipatory Cyclone & Infrastructure Cascade Intelligence Platform"

ISRO Asset	Key Capability	VAYU-RAKSHA Use Case
INSAT-3DS (Geostationary)	10-minute update cycle Dvorak cyclone intensity technique Cloud motion vectors	Near-real-time intensity tracking Cloud-top temperature for eyewall analysis Feed into VAYU Agent every 10 min
RISAT-1A (C-band SAR)	Through-cloud imaging 3m resolution Disaster mode activation	Ground truth flood maps under monsoon clouds Coastal infrastructure damage detection Validation of GEE surge model outputs
Oceansat-3 (OceanColor)	Sea Surface Temperature Ku-band scatterometer Ocean wind vectors	Bay of Bengal SST anomaly detection (warm water = intensification risk) Wind shear analysis for rapid intensification
Bhuvan/NDEM Geoportal	India infrastructure GIS NDEM emergency database Disaster response layers	Coastal district infrastructure nodes (admin) NDRF resource locations Shared real-time flood alerts
MOSDAC Portal	Satellite cyclone prediction NWP products Ocean state forecasts	Baseline cyclone track validation Wave height predictions for storm surge Complementary to IMD track data

GEE Satellite Processing Pipeline

GEE Dataset	Band / Product	Processing Step	Output
Sentinel-1 SAR GRD	VV + VH polarization	Speckle filter → thresholding → water mask	Flood extent raster (10m)
NASA NASADEM	Elevation (HGT)	Slope analysis + coastal zone delineation	Surge susceptibility map (30m)
GPM IMERG Final	precipitationCal (hourly)	72h accumulation → excess rainfall zone	Heavy flood risk layer
JRC Global Surface Water	Occurrence + seasonality	Baseline water mask for change detection	Flood change delta map
WorldPop	Population count (100m)	Intersection with inundation + cascade model	Affected population estimate
Dynamic World (10m)	Built-up + vegetation class	Infrastructure exposure classification	Built-up risk zone layer

[IMPLEMENTATION]

VAYU-RAKSHA

Anticipatory Cyclone & Infrastructure Case

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Build Sprint Plan (Submission by Sep 30)

Sprint 1 FOUNDATION	Days 1-2	LangGraph StateGraph wiring · CycloneState schema · GEE Python API setup · OSM infrastructure data pull · FastAPI skeleton · MOSDAC API credential setup
Sprint 2 CORE ENGINES	Days 3-4	Holland wind field implementation · Bathtub surge model · NetworkX cascade graph builder · Counterfactual optimizer (core algorithm) · Flood-penalized Dijkstra router
Sprint 3 AGENTS	Days 5-6	BHUMI Agent (GEE ingestion + SAR processing) · VAYU Agent (IMD + MOSDAC fusion + surge) · SETU Agent (cascade + routing) · SANCHAR Agent (Gemini multilingual + insurance)
Sprint 4 INTEGRATION	Day 7	NIRNAY Supervisor wiring · End-to-end test with Cyclone Fani historical data · React dashboard with MapboxGL cascade visualization · WebSocket live update
Sprint 5 POLISH + DEMO	Day 8	README + demo video · PDF submission materials · Cyclone Fani simulation demo · Performance tuning · GitHub cleanup · Final submission

TEAM: SYNTRIX | Build with AI — Code for Communities (2nd Edition)

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github.com/Niss54/vayu-raksha

[IMPACT]

VAYU-RAKSHA

Anticipatory Cyclone & Infrastructure

Impact Metric	Current State (India)	VAYU-RAKSHA Target	How
Pre-landfall action lead time	IMD gets advisory ~12h before landfall	Ranked action queue at T-72h	Cascade optimizer → action queue at T-72h ingestion
Infrastructure cascade prevention	Centralized manual coordination, partial coverage	Top 10 cascade initiators identified + ranked	Identified cascade graph + counterfactual optimizer
Population coverage per storm	Manual coordination, partial coverage	100% OSM-mapped assets auto-covered	GEE WorldPop × cascade graph coverage
Parametric insurance payout lag	Weeks loss adjustment	Hours: GEE satellite threshold verified	Sentinel-1 SAR + wind speed threshold trigger
Languages covered	IMD: English + regional pre-translation	6 languages (contextual, not just translated)	Gemini multilingual + cultural context prompting
ISRO satellite data used	No existing project uses INSAT-3DS	INSAT-3DS + RISAT-1A + Ocean Color Swath	MOSDAC API + Bhoonidhi disaster mode SAR

or optimizes PRE-LANDFALL hardening. VAYU-RAKSHA is the first 5-agent system to simulate cascade failure propagation and fuse ISRO MOSDAC + RISAT-1A with GEE.

✓ ONLY project with cascade failure graph

We model how infrastructure failures PROPAGATE — not just where they happen. This is what kills people.

Supervisor)

✓ ONLY project using ISRO native data

INSAT-3DS + RISAT-1A integration shows India-first thinking. Judges from GEE/Google will recognize this.

✓ ONLY project with counterfactual optimizer

Pre-landfall action ranking is what decision-makers actually need. Not another map.

✓ True 5-agent architecture

LangGraph StateGraph with typed state is genuinely novel vs. everyone using 'Gemini with tools'.

✓ End-to-end: satellite → action → insurance

Complete closed loop: from satellite data through AI reasoning to autonomous municipal dispatch to parametric payout.

✓ Reproducible + working code

Historical demo with Cyclone Fani (2019, Odisha) — real GEE data, real surge, real infrastructure overlay.

DEMO SCENARIO: We simulate Cyclone FANI (May 3, 2019 — Odisha landfall, 175 km/h winds). Using real IBTrACS track data + GEE Sentinel-1 SAR from that event + OSM Odisha infrastructure, VAYU-RAKSHA retrospectively demonstrates: (a) which cascade failures could have been prevented, (b) which hospitals needed pre-positioned fuel, and (c) what the parametric insurance trigger would have been — validating our model against real-world outcomes.

[FUTURE + TEAM]

VAYU-RAKSHA

Anticipatory Cyclone & Infrastructure

Phase 1 (Hackathon)	Odisha / AP coast demo · Cyclone Fani historical validation · Single-district cascade graph
Phase 2 (Post-hack)	Multi-state expansion (Odisha + AP + WB + TN) · NDRF API integration · Real-time IMD alert subscription · Data visualization
Phase 3 (Production)	Full Bay of Bengal coverage · Bangladesh/Myanmar APAC expansion · Real insurance API integration · NDMA integration
Phase 4 (Vision)	Regional APAC platform · Typhoon (Philippine Sea) + Cyclone (Arabian Sea) coverage · UN OCHA integration

Optimizes FIRE-LAND-FALL hardening. VAYU-RAKSHA is the first AI agent system to simulate cascade failure propagation and fuse ISRO MOSDAC + RISAT-1A with GEE.

- NDMA (National Disaster Management Authority) — real-time alert ingestion + advisory

BHUMI Agent (Dispatch)

- VAYU Agent

VAYU Agent

- SETU Agent

SETU Agent

- SANCHAR Agent

SANCHAR Agent

- NIRNAY (Supervisor)

NIRNAY (Supervisor)

- IRFC / Indian Reinsurance — parametric insurance trigger API for automated payouts

ISRO MOSDAC + RISAT-1A | Free Open Data

- Google Earth Engine | Gemini 3.7 Flash

Google Earth Engine | Gemini 3.7 Flash

Team Syntrix

Name	Role	Responsibilities
Nishant Maurya	Lead Developer & AI Architect	LangGraph multi-agent system · ISRO/GEE data pipeline · Cascade engine · GitHub: github.com/Niss54
Team Member 2	Backend & Geospatial	FastAPI · NetworkX graph · Storm surge model · GEE processing
Team Member 3	Frontend & Visualization	React dashboard · Mapbox GL · Cascade graph viz · Advisory PDF

READY TO BUILD?

GitHub Repo: github.com/Niss54/vayu-raksha

Stack: Python · LangGraph · GEE · Gemini 3.7 Flash · FastAPI · React · NetworkX

Demo: Cyclone FANI (2019 Odisha) — real data, real infrastructure, real cascade failure analysis

Contact: Build with AI — Code for Communities · Team Syntrix

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